Name : Tanishq Kumar Pandey || Class : MCADS11 || Roll No : 63 ||
Subject : NoSql and MongoDB

MongoDB Project

Write a Query for Each Question

1. Complex Filters & Projections

Q1. List the names and departments of students who have more than 85% attendance and are skilled in both "MongoDB" and "Python".

Answer1)

```
>_MONGOSH

> db.students.find(
    {
      attendance: { $gt: 85 },
      skills: { $all: ["MongoDB", "Python"] }
    },
    {
      _id: 0,
      name: 1,
      department: 1
    }
  ) // Name : Tanishq Kumar Pandey , Roll No : 63
```

Output :

No Output for the given query mean no data matched

Q2. Show all faculty who are teaching more than 2 courses. Display their names and the total number of courses they teach.

mongosh: localhost:27017

>_MONGOSH

```
> db.faculty.aggregate([
    {
      $project: {
        name: 1,
        totalCourses: { $size: "$courses" }
      }
    },
    {
      $match: {
        totalCourses: { $gt: 2 }
      }
    }
  ])// Name : Tanishq Kumar Pandey , Roll No : 63
```

Output :

```
< {
    _id: 'F029',
    name: 'Charles Newton',
    totalCourses: 3
  }
  {
    _id: 'F032',
    name: 'Julia Cole',
    totalCourses: 3
  }
  {
    _id: 'F040',
    name: 'Darrell Velasquez',
    totalCourses: 3
  }
  {
    _id: 'F048',
    name: 'Michael Poole',
    totalCourses: 3
  }
  {
    _id: 'F051',
    name: 'John Duran',
    totalCourses: 3
  }
  {
    _id: 'F061',
    name: 'Daniel Allen',
```

2. Joins ($lookup) and Aggregations

Q3. Write a query to show each student's name along with the course titles they are enrolled in (use $lookup between enrollments, students, and courses).

```
>_ mongosh: localhost:27017
>_MONGOSH
> db.enrollments.aggregate([
    {
      $lookup: {
        from: "students",
        localField: "student_id",
        foreignField: "_id",
        as: "student"
      }
    },
    {
      $lookup: {
        from: "courses",
        localField: "course_id",
        foreignField: "_id",
        as: "course"
      }
    },
    {
      $unwind: "$student"
    },
    {
      $unwind: "$course"
    },
    {
      $project: {
        _id: 0,
        studentName: "$student.name",
        courseTitle: "$course.title"
      }
    }
  ])// Name : Tanishq Kumar Pandey, Roll No: 63
```

Output:

```
< {
    studentName: 'Alexandra Bailey',
    courseTitle: 'Reactive neutral adapter'
  }
  {
    studentName: 'Megan Taylor',
    courseTitle: 'Sharable bifurcated paradigm'
  }
  {
    studentName: 'Alejandro Hart',
    courseTitle: 'Focused user-facing paradigm'
  }
  {
    studentName: 'Timothy Sparks',
    courseTitle: 'Focused user-facing paradigm'
  }
  {
    studentName: 'Juan Morris',
    courseTitle: 'Balanced asynchronous framework'
  }
  {
    studentName: 'Donna Morgan',
    courseTitle: 'Organic optimal product'
  }
  {
    studentName: 'Patricia Scott',
    courseTitle: 'Fully-configurable responsive solution'
  }
  {
    studentName: 'Carolyn Chandler',
    courseTitle: 'Horizontal attitude-oriented knowledgebase'
```

Q4. For each course, display the course title, number of students enrolled, and average marks (use $group).

```
>_ mongosh: localhost:27017
>_MONGOSH
> db.enrollments.aggregate([
    {
      $group: {
        _id: "$course_id",
        totalStudents: { $sum: 1 },
        avgMarks: { $avg: "$marks" }
      }
    },
    {
      $lookup: {
        from: "courses",
        localField: "_id",
        foreignField: "_id",
        as: "course"
      }
    },
    {
      $unwind: "$course"
    },
    {
      $project: {
        _id: 0,
        courseTitle: "$course.title",
        totalStudents: 1,
        avgMarks: { $round: ["$avgMarks", 2] }
      }
    }
  ])// Name : Tanishq Kumar Pandey , Roll No : 63
```

Output:

```
< {
    totalStudents: 1,
    courseTitle: 'Fully-configurable empowering data-warehouse',
    avgMarks: 64
  }
  {
    totalStudents: 2,
    courseTitle: 'Balanced national function',
    avgMarks: 68
  }
  {
    totalStudents: 1,
    courseTitle: 'Focused multi-state encoding',
    avgMarks: 67
  }
  {
    totalStudents: 1,
    courseTitle: 'Switchable regional open system',
    avgMarks: 67
  }
  {
    totalStudents: 2,
    courseTitle: 'Streamlined bandwidth-monitored structure',
    avgMarks: 77.5
  }
  {
    totalStudents: 1,
    courseTitle: 'Triple-buffered eco-centric implementation',
    avgMarks: 97
  }
```

3. Grouping, Sorting, and Limiting

Q5. Find the top 3 students with the highest average marks across all enrolled courses.

```
>_MONGOSH
> db.enrollments.aggregate([
    {
      $group: {
        _id: "$student_id",
        avgMarks: { $avg: "$marks" }
      }
    },
    {
      $sort: { avgMarks: -1 }
    },
    {
      $limit: 3
    },
    {
      $lookup: {
        from: "students",
        localField: "_id",
        foreignField: "_id",
        as: "student"
      }
    },
    {
      $unwind: "$student"
    },
    {
      $project: {
        _id: 0,
        studentName: "$student.name",
        avgMarks: { $round: ["$avgMarks", 2] }
      }
    }
```

```
      avgMarks: { $round: [ $avgMarks , 2] }
        }
      }
    ])// Name : Tanishq Kumar Pandey, Roll No : 63
< {
    studentName: 'Diane Phillips',
    avgMarks: 100
  }
  {
    studentName: 'Brandon Rios',
    avgMarks: 98
  }
  {
    studentName: 'Christopher Benson',
    avgMarks: 94
  }
```

Output:

```
< {
    studentName: 'Diane Phillips',
    avgMarks: 100
  }
  {
    studentName: 'Brandon Rios',
    avgMarks: 98
  }
  {
    studentName: 'Larry Ramsey',
    avgMarks: 94
  }
PROJECTMONGO > |
```

Q6. Count how many students are in each department. Display the department with the highest number of students.

```
>_ mongosh: localhost:27017

>_MONGOSH
> db.students.aggregate([
    {
      $group: {
        _id: "$department",
        totalStudents: { $sum: 1 }
      }
    },
    {
      $sort: { totalStudents: -1 }
    },
    {
      $limit: 1
    },
    {
      $project: {
        _id: 0,
        department: "$_id",
        totalStudents: 1
      }
    }
  ])// Name : Tanishq Kumar Pandey , Roll No: 63
< {
    totalStudents: 23,
    department: 'Electrical'
  }
PROJECTMONGO >
```

Output:

```
< {
    totalStudents: 23,
    department: 'Electrical'
  }
```

4. Update, Upsert, and Delete

Q7. Update attendance to 100% for all students who won any "Hackathon".

Input and Output:

```
>_ mongosh: localhost:27017

>_MONGOSH
> db.students.updateMany(
    {
      _id: {
        $in: db.activities.distinct("student_id", {
          type: "Hackathon",
          position: "Winner"
        })
      }
    },
    {
      $set: { attendance: 100 }
    }
  )// Name : Tanishq Kumar Pandey , Roll No : 63
< {
    acknowledged: true,
    insertedId: null,
    matchedCount: 0,
    modifiedCount: 0,
    upsertedCount: 0
  }
```

Q8. Delete all student activity records where the activity year is before 2022.

Input and Output :

```
>_ mongosh: localhost:27017

>_MONGOSH
> db.activities.deleteMany({
    year: { $lt: 2022 }
  })// Name : Tanishq Kumar Pandey, Roll No : 63
< {
    acknowledged: true,
    deletedCount: 0
  }
```

Q9. Upsert a course record for "Data Structures" with ID "C150" and credits 4—if it doesn't exist, insert it; otherwise update its title to "Advanced Data Structures".

Input and Output:

```
>_ mongosh: localhost:27017

>_MONGOSH
> db.courses.updateOne(
    { _id: "C150" },
    {
      $set: {
        title: "Advanced Data Structures",
        credits: 4
      }
    },
    {
      upsert: true
    }
  ) // Name: Tanishq Kumar Pandey , Roll NO : 63
< {
    acknowledged: true,
    insertedId: null,
    matchedCount: 1,
    modifiedCount: 0,
    upsertedCount: 0
  }
```

5. Array & Operator Usage

Q10. Find all students who have "Python" as a skill but not "C++".

```
>_ mongosh: localhost:27017
>_MONGOSH
>
> db.students.find({
    $and: [
      { skills: "Python" },
      { skills: { $nin: ["C++"] } }
    ]
  })// Name : Tanishq Kumar Pandey , Roll No : 63
```

Output:

```
< {
    _id: 'S004',
    name: 'Kyle Hale',
    dob: '2000-10-20',
    department: 'Electrical',
    skills: [
      'Python',
      'Java'
    ],
    attendance: 79.78
  }
  {
    _id: 'S008',
    name: 'Cody Whitehead',
    dob: '2003-11-25',
    department: 'Biotechnology',
    skills: [
      'JavaScript',
      'Python'
    ],
    attendance: 92.03
  }
```

```
    skills: [
      'Python',
      'AutoCAD'
    ],
    attendance: 96.64
  }
  {
    _id: 'S012',
    name: 'Steven Wong',
    dob: '2003-09-06',
    department: 'Biotechnology',
    skills: [
      'MongoDB',
      'Python'
    ],
    attendance: 77.17
  }
  {
    _id: 'S015',
    name: 'Cheryl Jackson',
    dob: '2006-04-20',
    department: 'Civil',
    skills: [
      'Research',
      'Python'
    ],
    attendance: 60.98
  }
  {
    _id: 'S023',
    name: 'Mr. Darius Newman',
```

```
    name: 'Mr. Darius Newman',
    dob: '2000-12-13',
    department: 'Mechanical',
    skills: [
      'Python',
      'SQL'
    ],
    attendance: 95.06
  }
  {
    _id: 'S031',
    name: 'Derrick Humphrey',
    dob: '2001-12-31',
    department: 'Civil',
    skills: [
      'Python',
      'Java'
    ],
    attendance: 69.29
  }
  {
    _id: 'S033',
    name: 'Paula Jenkins',
    dob: '2004-02-03',
    department: 'Biotechnology',
    skills: [
      'JavaScript',
      'Python'
    ],
    attendance: 99.8
```

```
}
{
  _id: 'S047',
  name: 'Barbara Jones',
  dob: '2003-12-27',
  department: 'Civil',
  skills: [
    'Python',
    'Research'
  ],
  attendance: 77.09
}
{
  _id: 'S053',
  name: 'Tracey Young',
  dob: '2006-05-26',
  department: 'Computer Science',
  skills: [
    'Python',
    'AutoCAD'
  ],
  attendance: 75.98
}
{
  _id: 'S064',
  name: 'Elizabeth Reed',
  dob: '2001-01-16',
  department: 'Computer Science',
  skills: [
    'Java',
    'Python'
```

```
  attendance: 95.53
}
{
  _id: 'S065',
  name: 'Brian Russell',
  dob: '2006-07-31',
  department: 'Mechanical',
  skills: [
    'Python',
    'Research'
  ],
  attendance: 92.29
}
{
  _id: 'S079',
  name: 'David Rivera',
  dob: '2000-05-29',
  department: 'Computer Science',
  skills: [
    'Python',
    'JavaScript'
  ],
  attendance: 70.67
}
{
  _id: 'S095',
  name: 'Taylor Webb',
  dob: '2003-05-04',
  department: 'Computer Science',
  skills: [
```

```
    department: 'Computer Science',
    skills: [
      'Linux',
      'Python'
    ],
    attendance: 82.91
  }
  {
    _id: 'S098',
    name: 'Erin Harris',
    dob: '2002-11-21',
    department: 'Biotechnology',
    skills: [
      'AutoCAD',
      'Python'
    ],
    attendance: 85.33
  }
  {
    _id: 'S099',
    name: 'Kyle Lee',
    dob: '2002-12-13',
    department: 'Mechanical',
    skills: [
      'Python',
      'JavaScript'
    ],
    attendance: 69.62
  }
PROJECTMONGO >
```

Q11. Return names of students who participated in "Seminar" and "Hackathon" both.

```
>_ mongosh: localhost:27017
>_MONGOSH
> db.Activities.aggregate([
    {
      $group: {
        _id: "$student_id",
        types: { $addToSet: "$type" }
      }
    },
    {
      $match: {
        types: { $all: ["Seminar", "Hackathon"] }
      }
    },
    {
      $lookup: {
        from: "students",
        localField: "_id",
        foreignField: "_id",
        as: "student"
      }
    },
    {
      $unwind: "$student"
    },
    {
      $project: {
        _id: 0,
        studentName: "$student.name"
      }
    }
  ]) // Name : Tanishq Kumar Pandey, Roll No : 63
```

Output:

```
< {
    studentName: 'Lydia Day'
  }
  {
    studentName: 'Taylor Webb'
  }
  {
    studentName: 'Carlos Bryant'
  }
  {
    studentName: 'Patricia Scott'
  }
  {
    studentName: 'Adam Solomon'
  }
PROJECTMONGO >
```

6. Subdocuments and Nested Conditions

Q12. Find students who scored more than 80 in "Web Development" only if they belong to the "Computer Science" department.

```
>_ mongosh: localhost:27017
>_MONGOSH
>
> db.enrollments.aggregate([
    {
      $lookup: {
        from: "students",
        localField: "student_id",
        foreignField: "_id",
        as: "student"
      }
    },
    {
      $lookup: {
        from: "courses",
        localField: "course_id",
        foreignField: "_id",
        as: "course"
      }
    },
    { $unwind: "$student" },
    { $unwind: "$course" },
    {
      $match: {
        "student.department": "Computer Science",
        "course.title": "Web Development",
        marks: { $gt: 80 }
      }
    },
    {
      $project: {
        _id: 0,
        studentName: "$student.name",
```

```
    $project: {
      _id: 0,
      studentName: "$student.name",
      marks: 1
    }
  }
]) // Name : Tanishq Kumar Pandey , Roll No : 63
```

Output:

No Output

7. Advanced Aggregation (Challenge Level)

Q13. For each faculty member, list the names of all students enrolled in their courses along with average marks per student per faculty.

```
>_ mongosh: localhost:27017
>_MONGOSH
> db.faculty.aggregate([
    {
      $lookup: {
        from: "courses",
        localField: "_id",
        foreignField: "faculty_id",
        as: "courses"
      }
    },
    { $unwind: "$courses" },

    {
      $lookup: {
        from: "enrollments",
        localField: "courses._id",
        foreignField: "course_id",
        as: "enrollments"
      }
    },
    { $unwind: "$enrollments" },

    {
      $lookup: {
        from: "students",
        localField: "enrollments.student_id",
        foreignField: "_id",
        as: "student"
      }
    },
    { $unwind: "$student" },
```

>_ mongosh: localhost:27017

```
>_MONGOSH
    { $unwind: "$student" },

    {
      $group: {
        _id: {
          facultyName: "$name",
          studentName: "$student.name"
        },
        avgMarks: { $avg: "$enrollments.marks" }
      }
    },

    {
      $group: {
        _id: "$_id.facultyName",
        students: {
          $push: {
            name: "$_id.studentName",
            avgMarks: { $round: ["$avgMarks", 2] }
          }
        }
      }
    },

    {
      $project: {
        _id: 0,
        facultyName: "$_id",
        students: 1
      }
    }
```

```
    {
      $project: {
        _id: 0,
        facultyName: "$_id",
        students: 1
      }
    }
  ]) // Name : Tanishq Kumar Pandey, Roll No : 63
```

Output:

```
< {
    students: [
      {
        name: 'Ryan Flores',
        avgMarks: 84
      },
      {
```

```
>_MONGOSH
      {
        name: 'Donna Spencer',
        avgMarks: 81
      }
    ],
    facultyName: 'Jacqueline Miller'
  }
  {
    students: [
      {
        name: 'Lydia Day',
        avgMarks: 92
      },
      {
        name: 'Kyle Lee',
        avgMarks: 97
      }
    ],
    facultyName: 'James Kirby'
  }
  {
    students: [
      {
        name: 'Steven Booth',
        avgMarks: 69
      },
      {
        name: 'Benjamin White',
        avgMarks: 59
      }
    ],
```

```
>_MONGOSH
      }
    ],
    facultyName: 'Ann Porter MD'
  }
  {
    students: [
      {
        name: 'Megan Taylor',
        avgMarks: 81
      },
      {
        name: 'Dr. Michael Griffin Jr.',
        avgMarks: 69
      }
    ],
    facultyName: 'Andrew Mcmahon'
  }
  {
    students: [
      {
        name: 'Colleen Todd',
        avgMarks: 52
      }
    ],
    facultyName: 'Ann Johnson'
  }
  {
    students: [
      {
        name: 'Nicholas Turner',
        avgMarks: 100
```

```
>_MONGOSH
          name: 'Nicholas Turner',
          avgMarks: 100
        },
        {
          name: 'Juan Morris',
          avgMarks: 80
        }
      ],
      facultyName: 'Thomas Lewis'
    }
    {
      students: [
        {
          name: 'Travis Johnson',
          avgMarks: 73
        },
        {
          name: 'Thomas Jackson',
          avgMarks: 85
        },
        {
          name: 'Marie Wilson',
          avgMarks: 61
        },
        {
          name: 'Christopher Benson',
          avgMarks: 94
        }
      ],
      facultyName: 'Kelly Huang'
```

```
>_MONGOSH
      ],
      facultyName: 'Kelly Huang'
    }
    {
      students: [
        {
          name: 'Marie Wilson',
          avgMarks: 89
        },
        {
          name: 'Patricia Scott',
          avgMarks: 84
        },
        {
          name: 'Andrew Wilson',
          avgMarks: 92
        }
      ],
      facultyName: 'Erin Allen'
    }
    {
      students: [
        {
          name: 'Daniel Brown',
          avgMarks: 75
        },
        {
          name: 'David Jones',
          avgMarks: 67
        }
      ],
```

```
>_MONGOSH
      facultyName: 'Kevin Murphy'
    }
    {
      students: [
        {
          name: 'Carolyn Chandler',
          avgMarks: 51
        }
      ],
      facultyName: 'Charles Cunningham'
    }
    {
      students: [
        {
          name: 'Reginald Oliver',
          avgMarks: 84
        }
      ],
      facultyName: 'Shelly Sawyer'
    }
    {
      students: [
        {
          name: 'Janet Jarvis',
          avgMarks: 67
        }
      ],
      facultyName: 'Laura Flores'
    }
    {
      students: [
```

```
>_MONGOSH
    students: [
      {
        name: 'Jessica Galvan',
        avgMarks: 64
      }
    ],
    facultyName: 'Kathryn Young'
  }
  {
    students: [
      {
        name: 'Bruce Blair',
        avgMarks: 78
      }
    ],
    facultyName: 'Jessica Black'
  }
  {
    students: [
      {
        name: 'Nicholas Turner',
        avgMarks: 51
      }
    ],
    facultyName: 'Autumn White'
  }
  {
    students: [
      {
        name: 'Monica Martin',
        avgMarks: 92
```

```
>_MONGOSH
  {
    students: [
      {
        name: 'Brian Russell',
        avgMarks: 82
      }
    ],
    facultyName: 'Darrell Velasquez'
  }
  {
    students: [
      {
        name: 'Timothy Sparks',
        avgMarks: 80
      },
      {
        name: 'Ryan Flores',
        avgMarks: 84
      }
    ],
    facultyName: 'Cassandra Diaz'
  }
  {
    students: [
      {
        name: 'Mary Bennett',
        avgMarks: 82
      }
    ],
    facultyName: 'Ashlee Russo'
  }
```

Q14. Show the most popular activity type (e.g., Hackathon, Seminar, etc.) by number of student participants.

Input and Output:

```
>_ mongosh: localhost:27017

>_MONGOSH

> db.Activities.aggregate([
    {
      $group: {
        _id: "$type",
        totalStudents: { $sum: 1 }
      }
    },
    {
      $setWindowFields: {
        sortBy: { totalStudents: -1 },
        output: {
          rank: { $rank: {} }
        }
      }
    },
    {
      $match: { rank: 1 }
    }
  ]) // Name : Tanishq Kumar Pandey, Roll No : 63
< {
    _id: 'Hackathon',
    totalStudents: 35,
    rank: 1
  }
  {
    _id: 'Seminar',
    totalStudents: 35,
    rank: 1
  }
```